

##(a)

```
#get data
setwd("/Users/apple/Documents/UOW/INFO911 Data Mining and Knowledge
Discovery/assignment1")
taxi_data <- read.csv("taxi.csv", header = TRUE, sep = ',', dec = '.')
```

```
install.packages("dbscan")
library(dbscan)
library(ggplot2)
```

Using the IQR approach for outlier detection is possible for the size of this dataset. However, outliers were not well identified using this method. The number of outliers was more than that of valid points and outliers concentrated in highly dense areas. Since the dataset is very large, using dbscan to detect outliers is not possible. Therefore, a representative sample is achieved by taking 100 observations from each DriveNo.

```
# Define the number of samples to take from each DriveNo
samples_per_driver <- 100
```

```
# Initialize an empty list for samples
sampled_list <- list()
```

```
# Get unique DriveNo values
driver_numbers <- unique(taxi_data$DriveNo)
```

```
# Stratified sampling
for (driver in driver_numbers) {
  driver_data <- taxi_data[taxi_data$DriveNo == driver, ]
  if (nrow(driver_data) >= samples_per_driver) {
    sampled_data <- driver_data[sample(nrow(driver_data), samples_per_driver), ]
    sampled_list[[as.character(driver)]] <- sampled_data
  }
}
```

```
# Combine all sampled data into a single data frame
stratified_sample <- do.call(rbind, sampled_list)
stratified_sample$status <- 'valid'
```

```
# Identify missing and invalid points
invalid_points <- is.na(stratified_sample$Latitude) | is.na(stratified_sample$Longitude) |
stratified_sample$Latitude < 41.8 | stratified_sample$Latitude > 42.05 |
stratified_sample$Longitude < 12.35 | stratified_sample$Longitude > 12.65
stratified_sample$status[invalid_points] <- 'invalid'
valid_points <- !invalid_points
```

```
# Apply DBSCAN to detect outliers
# Select relevant columns for clustering
coordinates <- stratified_sample[, c("Longitude", "Latitude")]
```

```

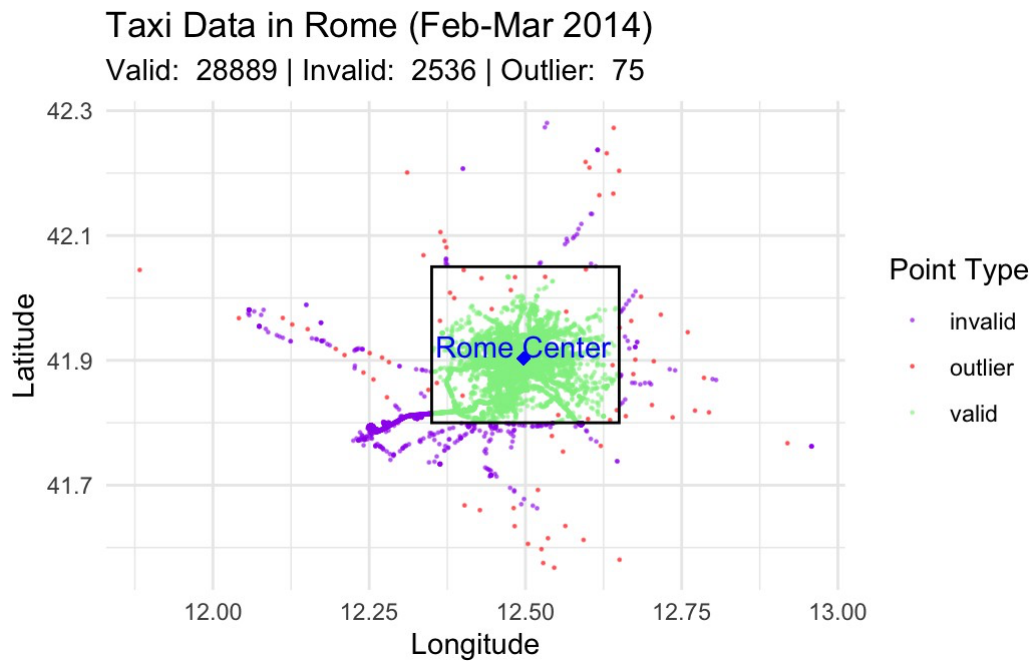
# Run DBSCAN
dbscan_result <- dbscan(coordinates, eps = 0.01, minPts = 2)

# Add cluster information to the sampled data
stratified_sample$cluster <- dbscan_result$cluster

# Identify outliers (cluster = 0)
outliers <- stratified_sample$cluster == 0
stratified_sample$status[outliers] <- 'outlier'
valid_points2 <- valid_points | !outliers

# Plot the sample
ggplot(stratified_sample, aes(x = stratified_sample$Longitude, y =
stratified_sample$Latitude, color = status)) +
  geom_point(alpha = 0.5, size=0.2) +
  scale_color_manual(values = c("valid" = "lightgreen",
                                "outlier" = "red",
                                "invalid" = "purple")) +
  labs(title = "Taxi Data in Rome (Feb-Mar 2014)",
        subtitle = paste("Valid: ", table(stratified_sample$status)['valid'],
                          "| Invalid: ", table(stratified_sample$status)['invalid'],
                          "| Outlier: ", table(stratified_sample$status)['outlier']),
        x = "Longitude",
        y = "Latitude",
        color = "Point Type") +
  theme_minimal() +
  coord_fixed(ratio = 1) +
  # Add a point for Rome's center
  geom_point(aes(x = rome_center_long, y = rome_center_lat),
             color = "blue", size = 2, shape = 18) +
  annotate("text", x = rome_center_long, y = rome_center_lat + 0.02,
          label = "Rome Center", color = "blue") +
  #Add a boundary
  annotate("rect", xmin = 12.35, xmax = 12.65, ymin = 41.8, ymax = 42.05, color = 'black',
          fill=NA)

```



```
#remove outliers and invalid points
```

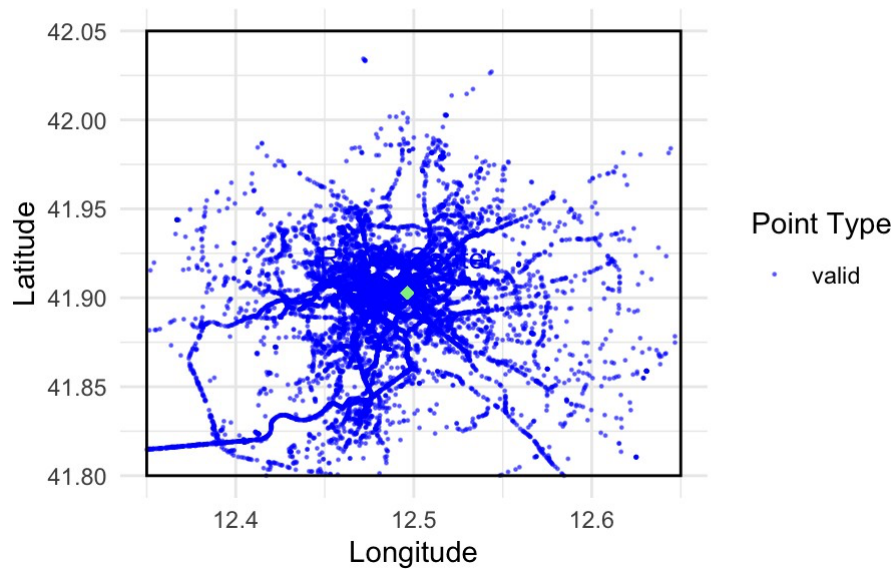
```
cleaned_data <- subset(stratified_sample, status != "outlier" & status != "invalid")
```

Plot cleaned data

```
ggplot(cleaned_data, aes(x = cleaned_data$Longitude, y = cleaned_data$Latitude, color =
status)) +
  geom_point(alpha = 0.5, size=0.2) +
  scale_color_manual(values = c("valid" = "blue",
                                "outlier" = "red",
                                "invalid" = "purple")) +
  labs(title = "Cleaned Taxi Data in Rome (Feb-Mar 2014)",
        subtitle = paste("Valid: ", table(cleaned_data$status)['valid'],
                          "| Invalid: ", table(cleaned_data$status)['invalid'],
                          "| Outlier: ", table(cleaned_data$status)['outlier']),
        x = "Longitude",
        y = "Latitude",
        color = "Point Type") +
  theme_minimal() +
  coord_fixed(ratio = 1) +
  # Add a point for Rome's center
  geom_point(aes(x = rome_center_long, y = rome_center_lat),
             color = "lightgreen", size = 2, shape = 18) +
  annotate("text", x = rome_center_long, y = rome_center_lat + 0.02,
          label = "Rome Center", color = "blue") +
  #Add a boundary
  annotate("rect", xmin = 12.35, xmax = 12.65, ymin = 41.8, ymax = 42.05, color = 'black',
          fill=NA)
```

Cleaned Taxi Data in Rome (Feb-Mar 2014)

Valid: 28889 | Invalid: NA | Outlier: NA



Compute the minimum, maximum, and mean location values

```
min_lat <- min(cleaned_data$Latitude) #41.80004
max_lat <- max(cleaned_data$Latitude) #42.03442
mean_lat <- mean(cleaned_data$Latitude) #41.89835
```

```
min_long <- min(cleaned_data$Longitude) #12.35006
max_long <- max(cleaned_data$Longitude) #12.64675
mean_long <- mean(cleaned_data$Longitude) #12.48462
```

Obtain the most active, least active, and average active taxi drivers

#convert datetime to R format

```
cleaned_data$Date.and.Time <- as.POSIXct(cleaned_data$Date.and.Time)
```

#find Max and Min time value points for each driver

```
time_summary <- aggregate(Date.and.Time ~ DriveNo, data = cleaned_data,
                           FUN = function(x) c(Max = max(x), Min = min(x)))
```

#the function returns a matrix of max and min time value points

```
str(time_summary)
```

#we need to convert the matrix to the dataframe format for further retrievals and calculations

```
time_summary <- data.frame(DriveNo = time_summary$DriveNo,
                           Date.and.Time.Max = time_summary$Date.and.Time[, "Max"],
                           Date.and.Time.Min = time_summary$Date.and.Time[, "Min"])
```

#convert datetime to R format

```
time_summary$Date.and.Time.Max <- as.POSIXct(time_summary$Date.and.Time.Max)
```

```
time_summary$Date.and.Time.Min <- as.POSIXct(time_summary$Date.and.Time.Min)
```

#calculate the amount of time driven for each driver

```
time_summary$Time.Difference <- time_summary$Date.and.Time.Max -  
time_summary$Date.and.Time.Min
```

```
most_active <- max(time_summary$Time.Difference) #Time difference of 43145.44 mins  
least_active <- min(time_summary$Time.Difference) #Time difference of 13.80055 mins  
average_active <- mean(time_summary$Time.Difference) #Time difference of 37256.31  
mins
```

Driver147 analytics

```
# Get data
```

```
setwd("/Users/apple/Documents/UOW/INFO911 Data Mining and Knowledge  
Discovery/assignment1")
```

```
student_data <- read.table("Student_Taxi_Mapping.txt", sep=" ", header = FALSE)  
colnames(student_data) <- c("SID", "Taxi_ID")
```

```
# Retrieve the taxi code
```

```
mycode <- 8111
```

```
taxi_code <- student_data[student_data$SID == mycode, ][2]
```

Plot the location points for taxi=ID

```
# The data plot for driver 147 is generated from all rows that contain DriveNo = 147 from the  
full dataset because using 100 observations of DriveNo147 from the stratified sample makes  
the dataset for driver147 too small
```

```
# Create a dataframe containing data for driver 147
```

```
taxi_data <- read.csv('taxi.csv', header = TRUE, sep = ',', dec = '.')
```

```
driver_147 <- taxi_data[taxi_data$DriveNo == 147, ]
```

```
driver_147$status <- 'valid'
```

```
# Identify missing and invalid points
```

```
invalid_points147 <- is.na(driver_147$Latitude) | is.na(driver_147$Longitude) |
```

```
driver_147$Latitude < 41.8 | driver_147$Latitude > 42.05 |
```

```
driver_147$Longitude < 12.35 | driver_147$Longitude > 12.65
```

```
driver_147$status[invalid_points147] <- 'invalid'
```

```
valid_points147 <- !invalid_points147
```

```
# Apply DBSCAN to detect outliers
```

```
# Select relevant columns for clustering
```

```
coordinates147 <- driver_147[, c("Longitude", "Latitude")]
```

```
# Run DBSCAN
```

```
dbscan_result147 <- dbscan(coordinates147, eps = 100, minPts = 10)
```

```
# Add cluster information to the sampled data
```

```
driver_147$cluster <- dbscan_result147$cluster
```

```
table(dbscan_result147$cluster)
```

```
# Identify outliers (cluster = 0)
```

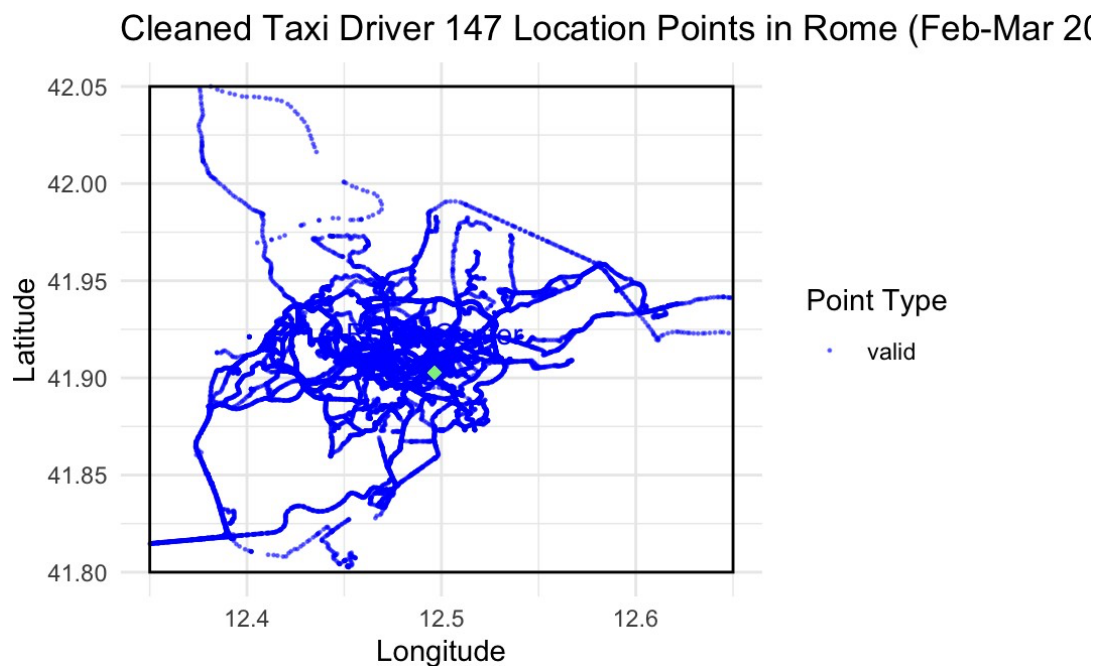
```

outliers147 <- driver_147$cluster == 0
driver_147$status[outliers147] <- 'outlier'
valid_points147_2 <- valid_points147[!outliers147]

# Plot cleaned dataset fo driver 147
cleaned_data147 <- subset(driver_147, status != "outlier" & status != "invalid")

ggplot(cleaned_data147, aes(x = cleaned_data147$Longitude, y =
cleaned_data147$Latitude, color = status)) +
  geom_point(alpha = 0.5, size=0.2) +
  scale_color_manual(values = c("valid" = "blue",
                                "outlier" = "red",
                                "invalid" = "purple")) +
  labs(title = "Taxi Driver 147 Location Points in Rome (Feb-Mar 2014)",
        x = "Longitude",
        y = "Latitude",
        color = "Point Type") +
  theme_minimal() +
  coord_fixed(ratio = 1) +
  # Add a point for Rome's center
  geom_point(aes(x = rome_center_long, y = rome_center_lat),
             color = "lightgreen", size = 2, shape = 18) +
  annotate("text", x = rome_center_long, y = rome_center_lat + 0.02,
          label = "Rome Center", color = "blue") +
  #Add a boundary
  annotate("rect", xmin = 12.35, xmax = 12.65, ymin = 41.8, ymax = 42.05, color = 'black',
fill=NA)

```



Compare the mean, min, and max location value of taxi=ID with the global mean, min, and max.

```
min_lat147 <- min(cleaned_data147$Latitude) #41.8026
max_lat147 <- max(cleaned_data147$Latitude) #42.05
mean_lat147 <- mean(cleaned_data147$Latitude) #41.91422

min_long147 <- min(cleaned_data147$Longitude) #12.35006
max_long147 <- max(cleaned_data147$Longitude) #12.64925
mean_long147 <- mean(cleaned_data147$Longitude) #12.45722
```

Compare total time driven by taxi=ID with the global mean, min, and max values.

```
time_driven147 <- max(cleaned_data147$Date.and.Time) -
min(cleaned_data147$Date.and.Time)
hours_driven147 <- as.numeric(time_driven147, units = "hours") #602.2279 hours
#Because the sample only consists of 31500 samples, it does not well represent the original
dataset. That explains why these metrics of driver 147 and those of the whole dataset are so
different.
```

Compute the distance traveled by taxi=ID

```
R <- 6371000
dlon147 <- max_long147 - min_long147
dlat147 <- max_lat147 - min_lat147
a147 <- (sin(dlat147/2))^2 + cos(min_lat147) * cos(max_lat147) * (sin(dlon147/2))^2
c147 = 2 * a147*tan2(sqrt(a147), sqrt(1-a147))
distance_travelled147 <- R * c147 #1795093
```